

Estimating the basic reproduction number post-pandemic

I am implementing the methodology proposed by Nick during the SISIMID 2024 course to estimate the basic reproductive number $\mathcal{R}_0$ for the first wave of the 2009 influenza pandemic in both the United States (US) and Mexico. This approach combines epidemiological data with a Bayesian inference framework to obtain posterior distributions for $\mathcal{R}_0$, allowing for the quantification of uncertainty and the comparison of transmission potential between the two countries during the early phase of the outbreak.

In this work we do not model under-reporting.

First, we use the below theorem.

Teorema 0.1. For a given epidemic outbreak that follows a SIR model. The fracción π (attack rate) of infected people at the end of the outbreak is

$$\pi = 1 - \exp(-\mathcal{R}_0\pi). \quad (1)$$

Let us denote with Y the total infected people at the end of the outbreak in a population of size N . Then, we define the expected value of Y as

$$\mathbb{E}[Y] := N\pi \quad (2)$$

Note: we assume that π the attack rate is the probability of infection in the population.

1. Data and model

For each country, we observe the total number of infected individuals in each state, denoted by Y_k , where $k = 1, 2, \dots, n$ indexes the states. The corresponding population size of state k is denoted by N_k .

We assume that the observed counts Y_k follow a Negative Binomial distribution, which allows for overdispersion relative to the Poisson model. Specifically, we write

$$Y_k \sim \text{NegBinom}\left(\frac{r}{\mu_k + r}, r\right), \quad (3)$$

where $r > 0$ is the dispersion parameter, and the expected number of infections in state k is

$$\mu_k = N_k \pi.$$

The probability mass function of Y_k is

$$\mathbb{P}[Y_k = y_k] = \frac{\Gamma(y_k + r)}{y_k! \Gamma(r)} \left(\frac{\mu_k}{\mu_k + r}\right)^{y_k} \left(\frac{r}{\mu_k + r}\right)^r. \quad (4)$$

This parametrization ensures that

$$\mathbb{E}[Y_k] = \mu_k, \quad \text{Var}(Y_k) = \mu_k + \frac{\mu_k^2}{r}.$$

The log-likelihood function for the parameter (π, r) is in proportion to

$$L(\pi, r) \propto \sum_{k=1}^n \left(\log(\Gamma(y_k + r)) - \log(\Gamma(r)) + y_k \log\left(\frac{\mu_k}{\mu_k + r}\right) + r \log\left(\frac{r}{r + \mu_k}\right) \right)$$

2. Bayesian inference

We apply bayesian inference with a non-informative prior for the dispersion parameter r and with a beta distribution as a prior for the parameter π . This is

$$\text{Beta}(\pi, a, b) = \frac{\Gamma(a + b)}{\Gamma(a)\Gamma(b)} \pi^{a-1} (1 - \pi)^{b-1}.$$

So, the posterior is proportional to

$$\mathbb{P}(\pi, r | Y_1, Y_2, \dots, Y_n) \propto L(\pi, r) + (a - 1) \log(\pi) + (b - 1) \log(1 - \pi).$$

3. Numerical results

Figures 1 and 2 show the contour plots of the log-likelihood function for the parameters (π, r) in each country. In both cases, the optimum is achieved in the interior of the parameter space. For Mexico, however, it is more difficult to visually identify the optimum due to the relatively flat shape of the likelihood surface.

When implementing the Metropolis–Hastings method, I used the hyper-parameters $a = 20$ and $b = 40$ for the Beta prior distribution of π . Under this parametrization, the prior mean of π is given by

$$\mathbb{E}[\pi] = \frac{a}{a+b} = \frac{20}{60} = \frac{1}{3},$$

and its standard deviation is

$$\text{sd}(\pi) = \sqrt{\frac{ab}{(a+b)^2(a+b+1)}} = 0.06.$$

Using the transformation

$$\mathcal{R}_0 = \frac{-\log(1-\pi)}{\pi},$$

the prior mean of $\mathcal{R}_0$ corresponds approximately to $\mathcal{R}_0 = 1.22$.

Figures 6 and 3 show the comparison of the prior and posterior marginal distributions for the basic reproductive number estimates of each country.

Figure 7 and 4 show the predictive posteriors for the data.

4. Conclusion

Using a prior distribution is important in our inference. when is not the case, the marginal posterior for $\mathcal{R}_0$ is very close to 1 with tiny variation, see the contour plot for each log-likelihood function.

Next steps:

- make a summary of the predictions (media, and 95 % bounds)
- what prior is best?
- how do we incorporate the under-reporting in the data?

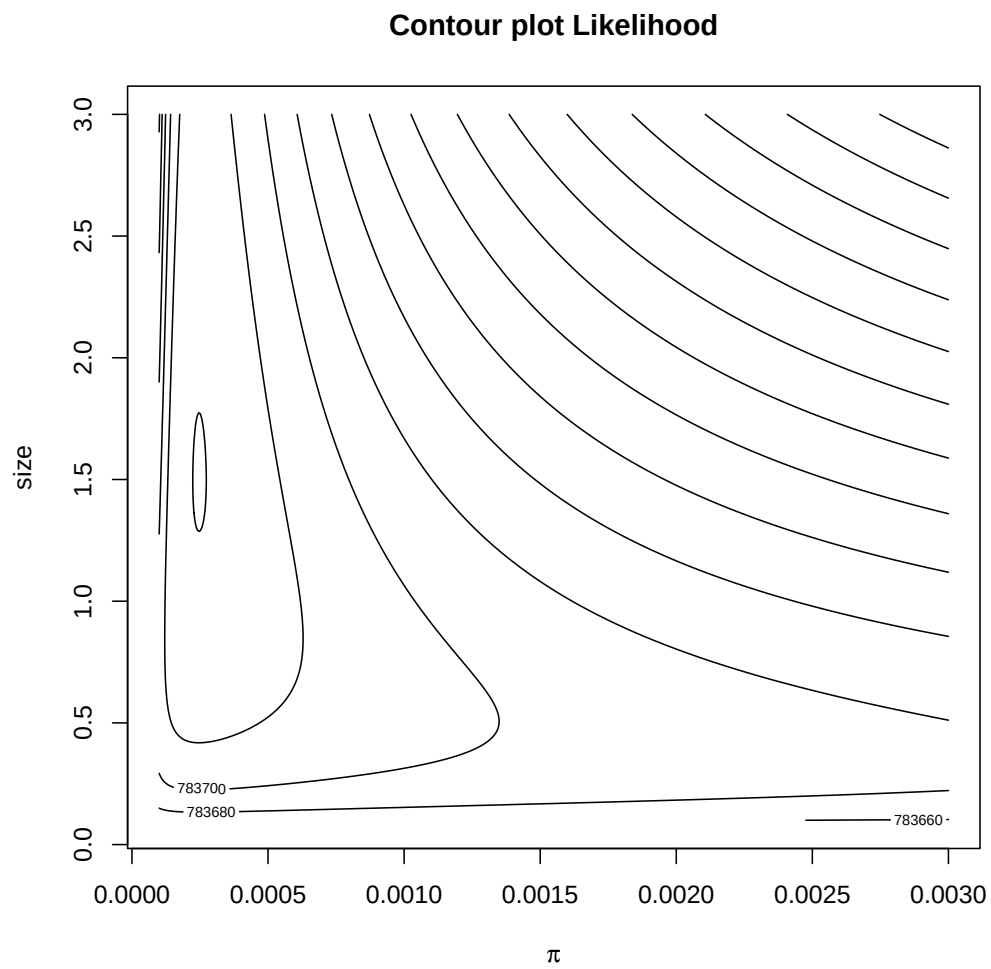


Figure 1: Contour plot for the loglikelihood function using data from US.

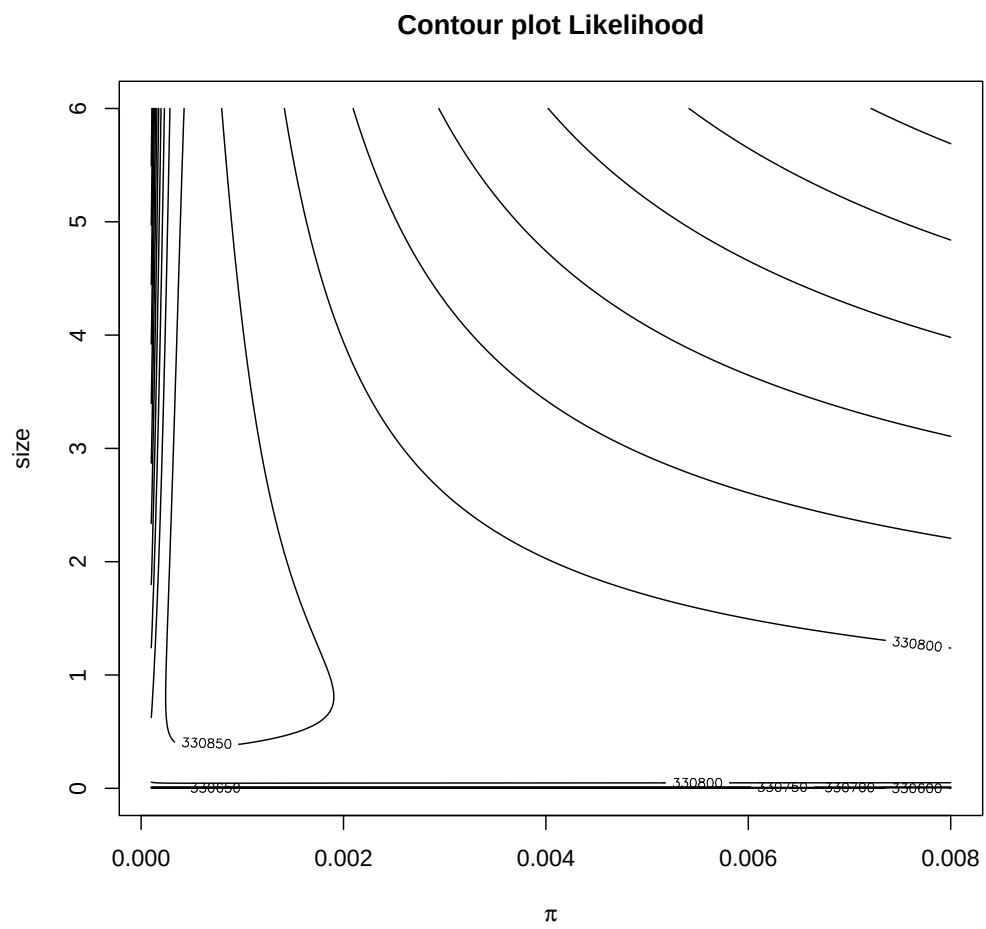


Figura 2: MX loglikelihood funtion

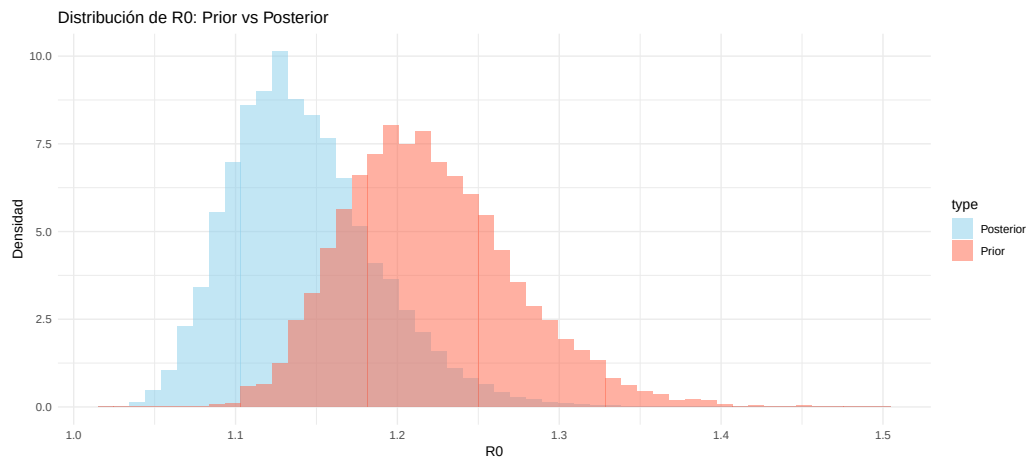


Figura 3: Comparison of the prior and marginal posterior for the basic reproduction number $\mathcal{R}_0$ for United States. Posterior median $\mathcal{R}_0 \approx 1.14$ with 95 % credible interval (1.07, 1.24).

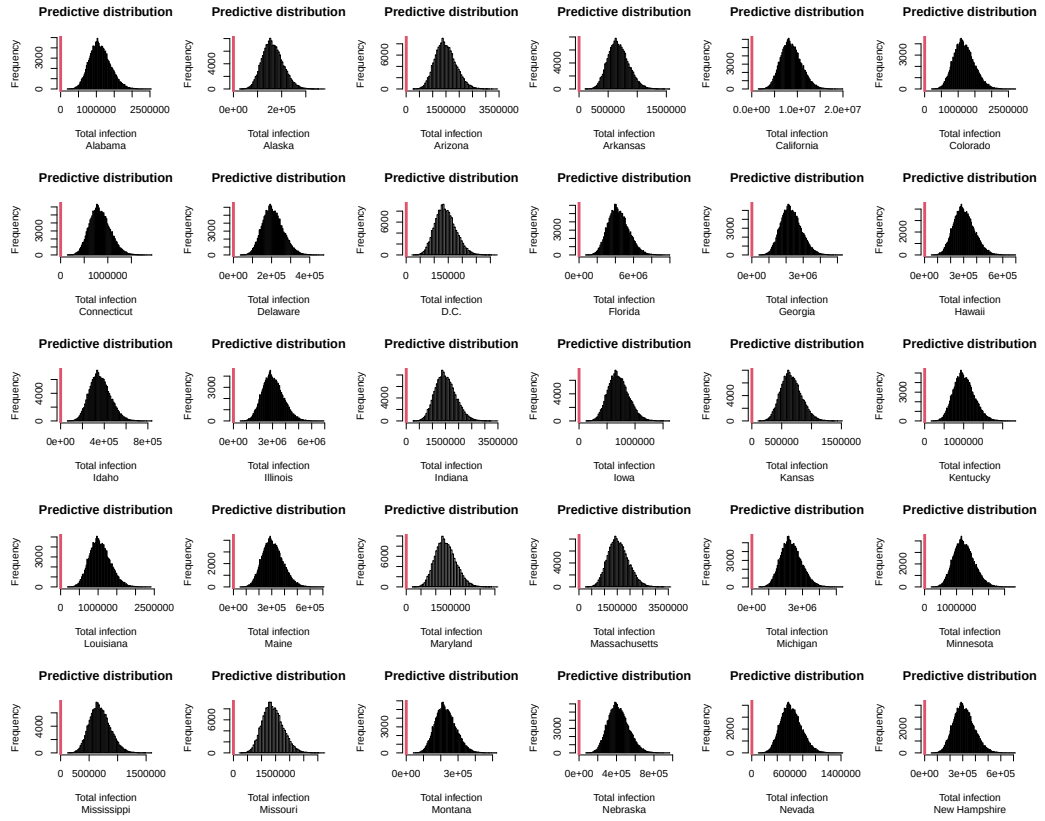


Figure 4: Redline: total cases reported for United States states. According to the predictive distributions, the probability of all observed data are very small (zero) and they are under-estimated.

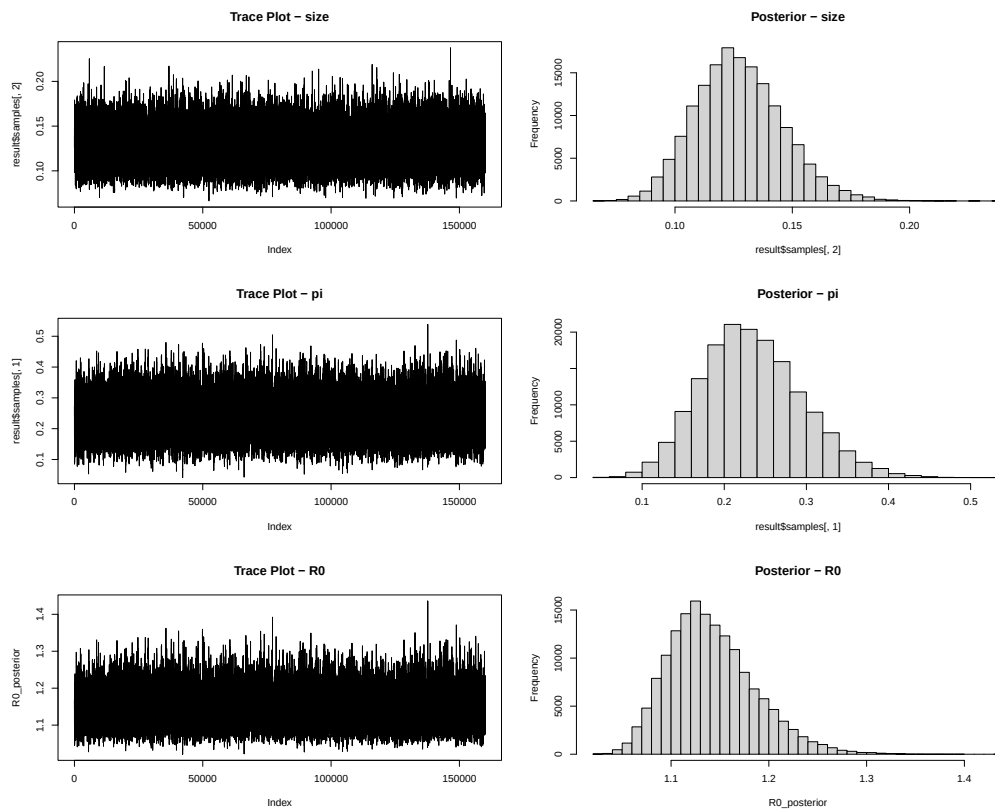


Figure 5: Marginal posterior for United States parameters

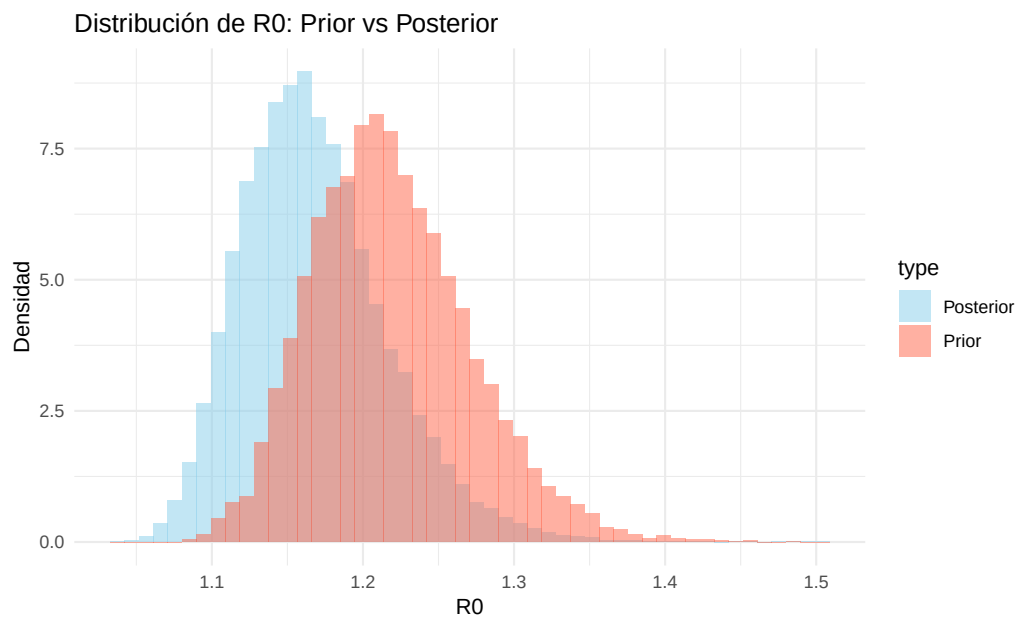


Figura 6: Comparison of the prior and marginal posterior for the basic reproduction number $\mathcal{R}_0$ for Mexico. Posterior median $\mathcal{R}_0 \approx 1.16$ with 95 % credible interval (1.09, 1.28).

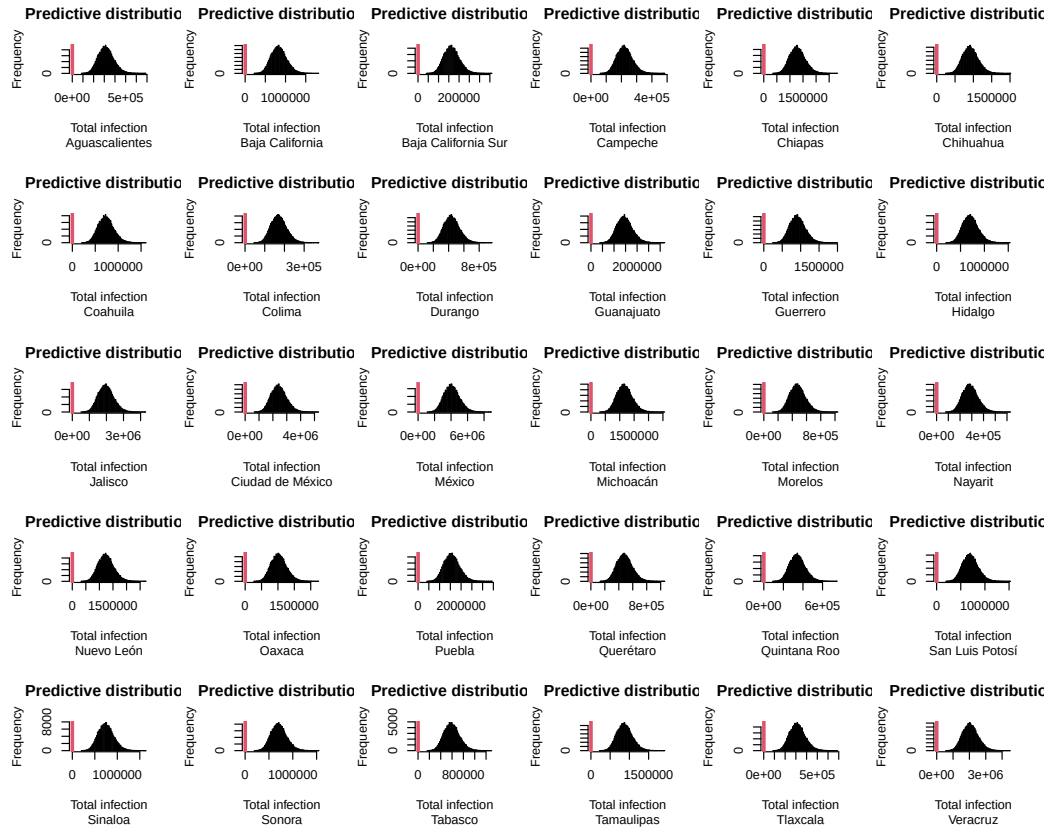


Figura 7: Redline: total cases reported for Mexican states. According to the predictive distributions, the probability of all observed data are very small (zero) and they are under-estimated.

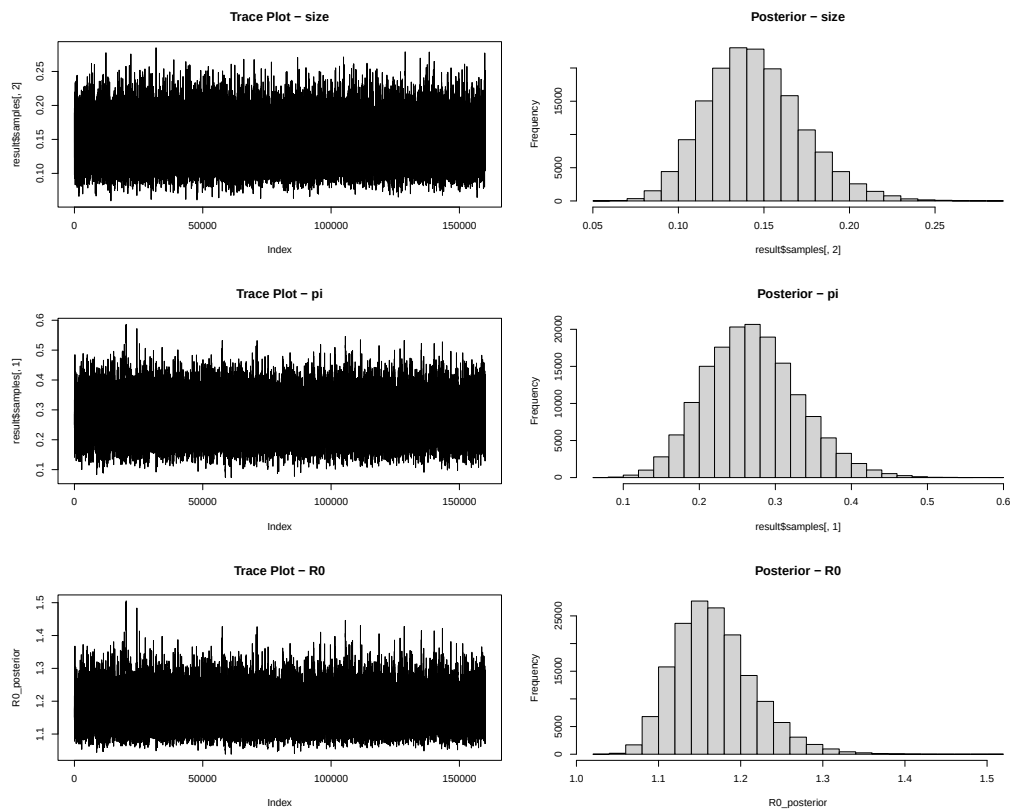


Figura 8: Marginal posterior for Mexican parameters